LAB 01

Installing Ubuntu Server on VMware Workstation Pro and Connecting with SSH

Name	Eman Fatima
Registration Number	2023-BCS-025
Course Code	[COURSE CODE]
Section	A
GitHub Username	Eman-Fatima-7125

Task 1: Check Available Host Storage

Explanation: A virtual machine stores its virtual disk as a file on the host computer, so the Windows drive must have enough free space before the lab starts. I opened Settings, then System, Storage and Disks & volumes, and checked drive C:. It has 237 GB in total with 34.3 GB free, which is more than enough for the 20 GB virtual disk used for Ubuntu Server.

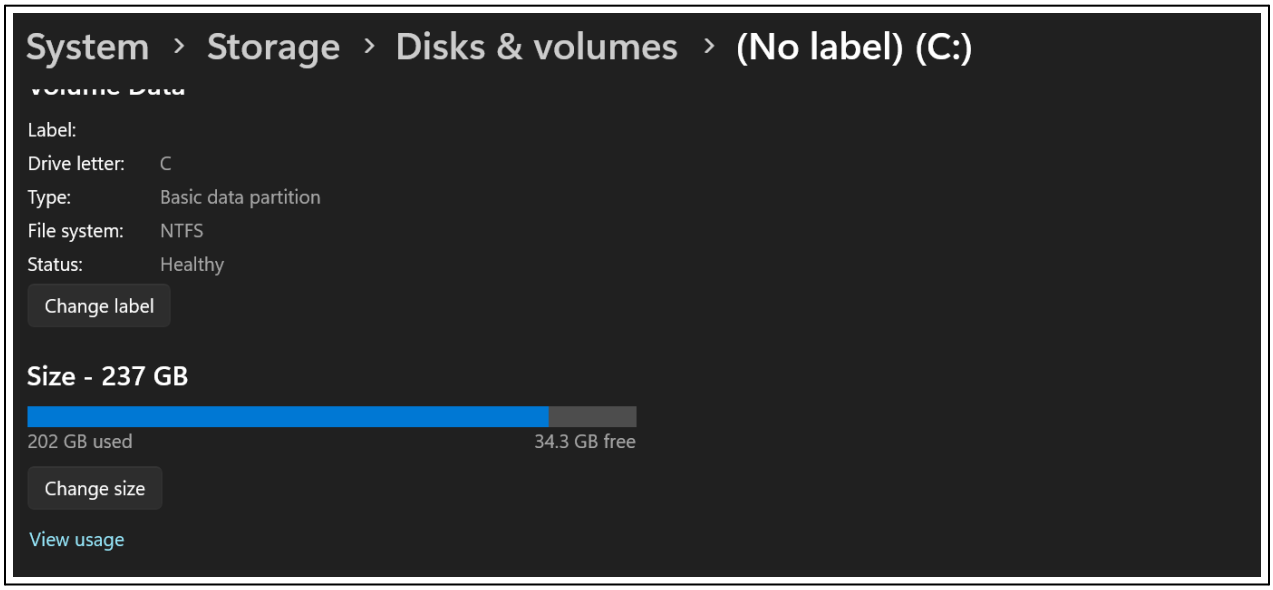


Figure 1.1: Windows Disks & volumes page for drive C: showing 237 GB total size and 34.3 GB free, confirming enough space for the Ubuntu Server virtual disk.

Task 2: GitHub Profile with Actual Name

Explanation: The GitHub account identifies me throughout this course and must show my complete real name. I opened Profile settings and confirmed that the Name field contains my full name, and that the account username is Eman-Fatima-7125. This username is also reused as the Ubuntu username later in the lab.

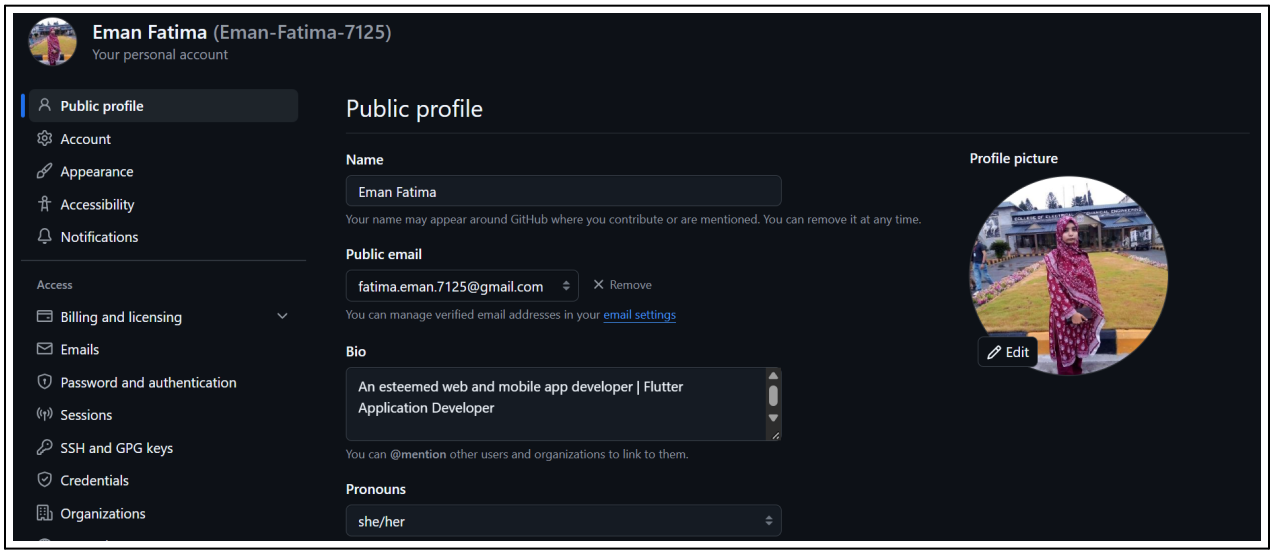


Figure 2.1: GitHub public profile showing the name Eman Fatima and the username Eman-Fatima-7125.

Task 3: Student Marks Portal Sign-in and Enrollment

Explanation: The Student Marks Portal records my academic progress, and it must use the same GitHub account as the rest of the lab. I opened the portal, chose Continue with GitHub, and authorized the sign-in. The Academic Record page then showed my GitHub-authenticated profile, with my registration number, GitHub username, email address and one approved course.

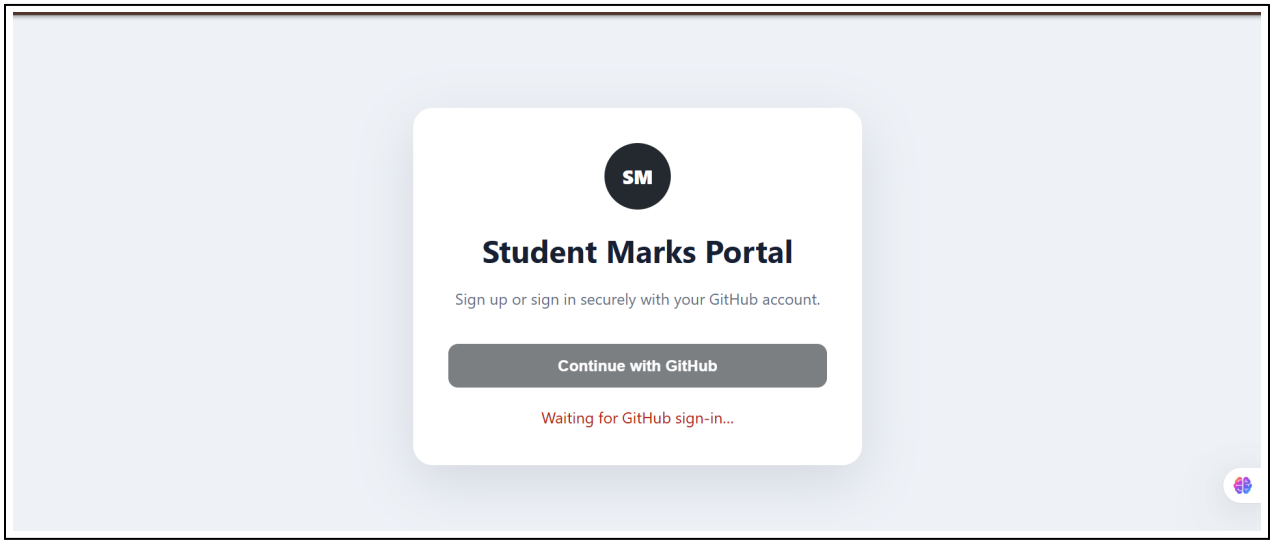


Figure 3.1: Student Marks Portal sign-in page with the Continue with GitHub option.

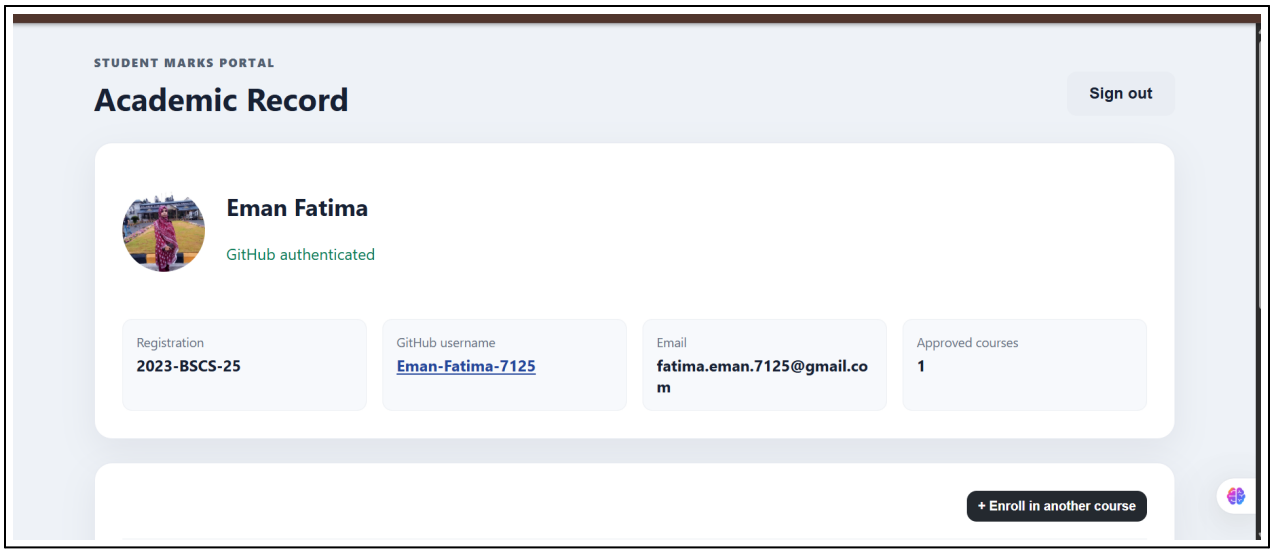


Figure 3.2: Portal Academic Record page showing GitHub authenticated, registration, GitHub username Eman-Fatima-7125, email, and one approved course (also covers the enrollment-submitted evidence).

Task 4: Download the Ubuntu Server ISO

Explanation: An ISO file is a disc image that contains the operating system installer. I downloaded the Ubuntu Server live installer, ubuntu-26.04.1-live-server-amd64.iso, to my Windows computer. VMware later uses this file as a virtual installation disc, so no physical DVD or USB drive is needed.

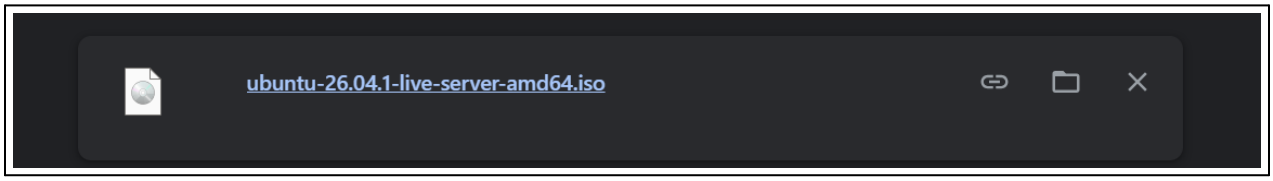


Figure 4.1: Downloaded installer file ubuntu-26.04.1-live-server-amd64.iso.

Task 5: Open VMware Workstation Pro

Explanation: VMware Workstation Pro is the virtualization software that runs Ubuntu Server as a guest system inside Windows. I launched VMware Workstation 16 Pro and reached its Home tab, which offers the Create a New Virtual Machine option used in the next task.

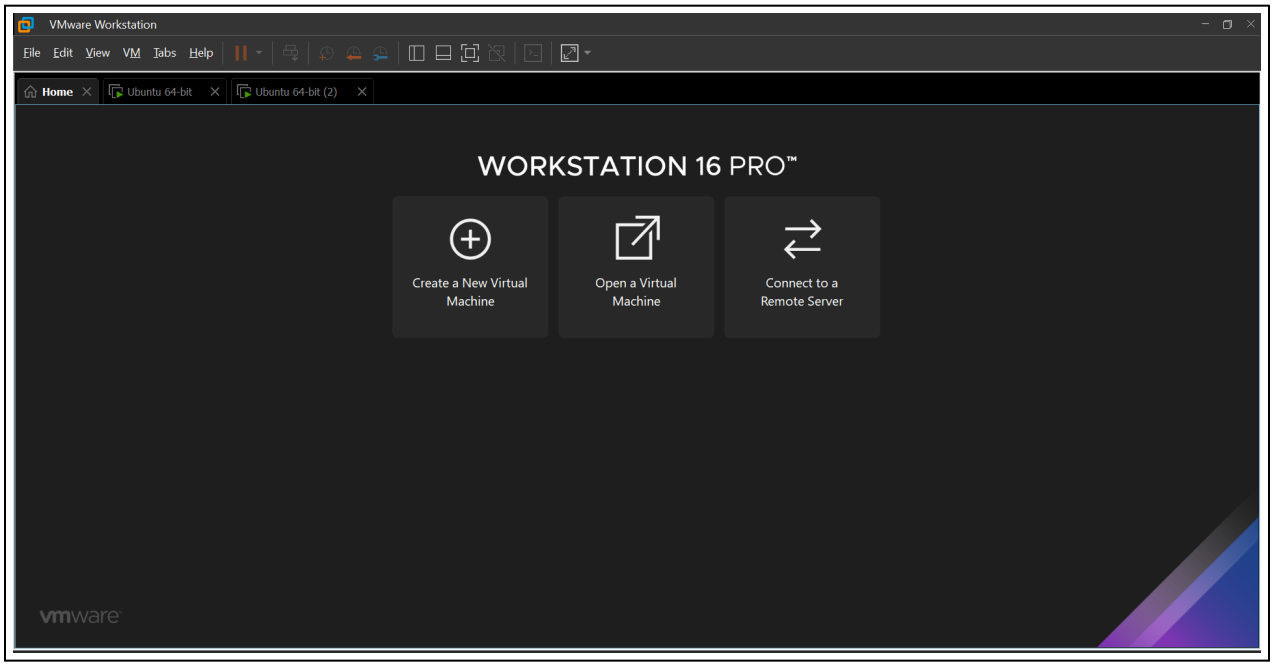


Figure 5.1: VMware Workstation 16 Pro home screen with the Create a New Virtual Machine option.

Task 6: Start the New Virtual Machine Wizard

Explanation: The New Virtual Machine Wizard guides the creation of the virtual machine step by step. I selected the Typical (recommended) configuration, which sets sensible hardware defaults for the guest system without the extra options of the Custom (advanced) mode.

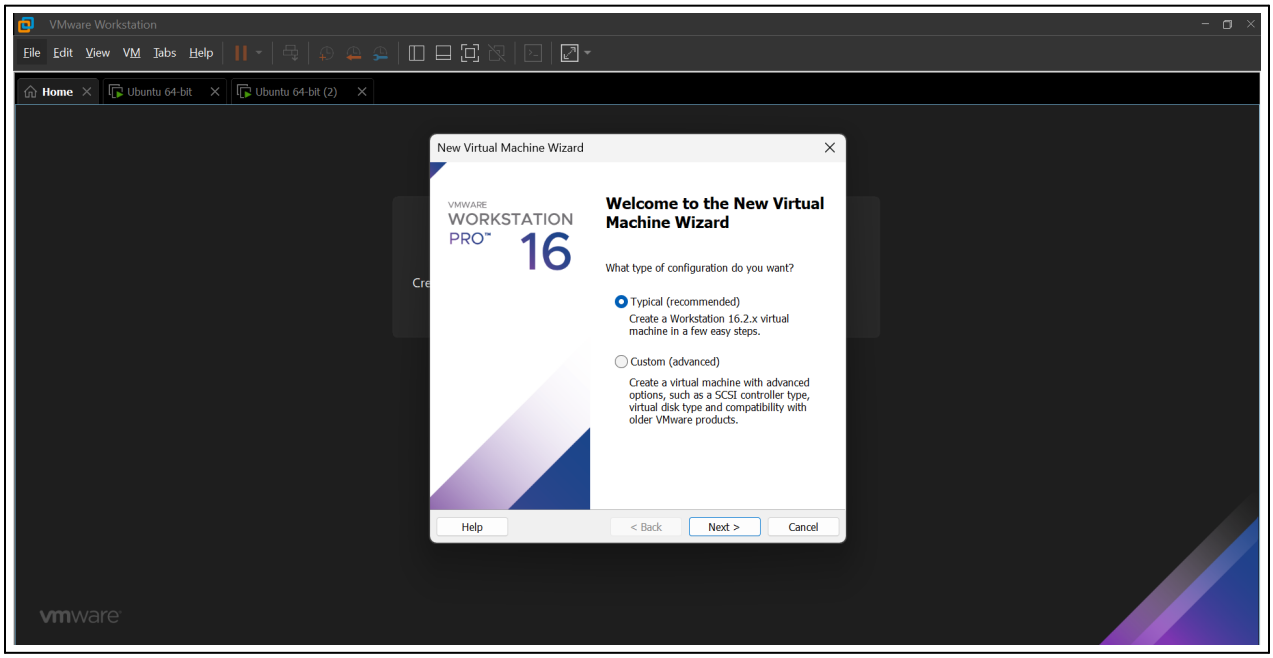


Figure 6.1: New Virtual Machine Wizard with Typical (recommended) configuration selected (covers the VM creation wizard and typical configuration evidence).

Task 7: Select the Ubuntu ISO

Explanation: The wizard asks how the guest operating system will be installed. I chose Installer disc image file (iso) and browsed to the Ubuntu Server ISO downloaded in Task 4. VMware inspected the file and reported that it detected Ubuntu 64-bit Server 26.04.1, which shows that the correct image was loaded and that Easy Install is available.

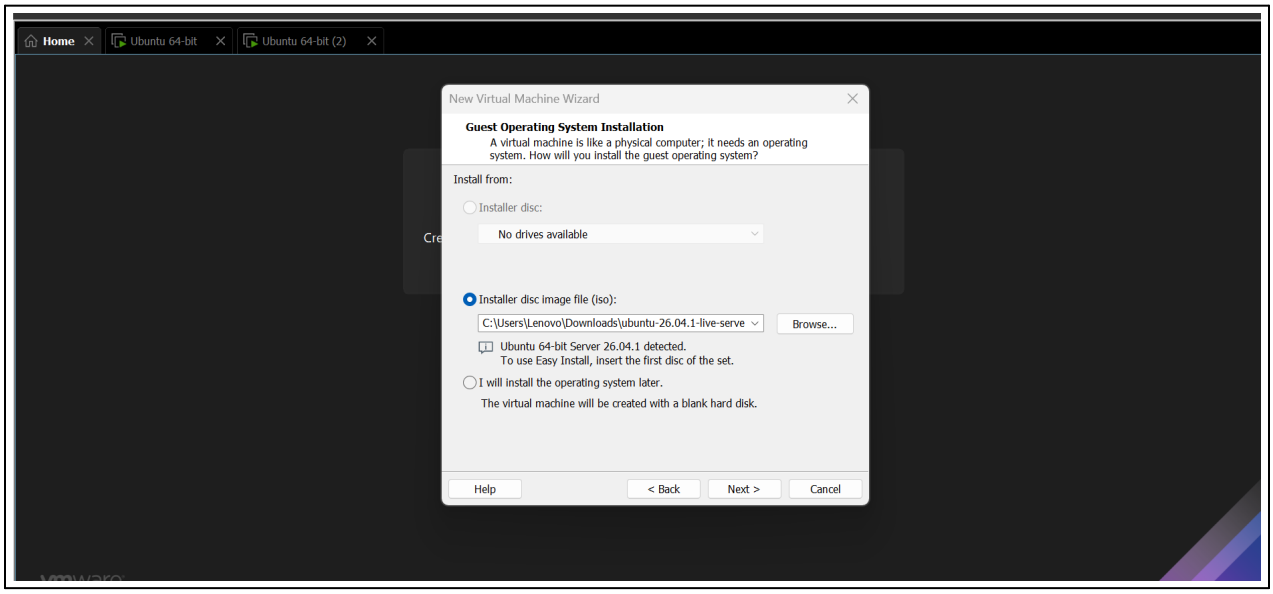


Figure 7.1: Installer disc image file set to the Ubuntu 26.04.1 live server ISO; the wizard detected Ubuntu 64-bit Server 26.04.1.

Task 8: Installer Language Selection

Explanation: After the virtual machine was powered on, it booted from the ISO and the Ubuntu Server installer started. The first screen asks for the language of the installer and of the installed system. I selected English using the arrow keys and the Enter key.

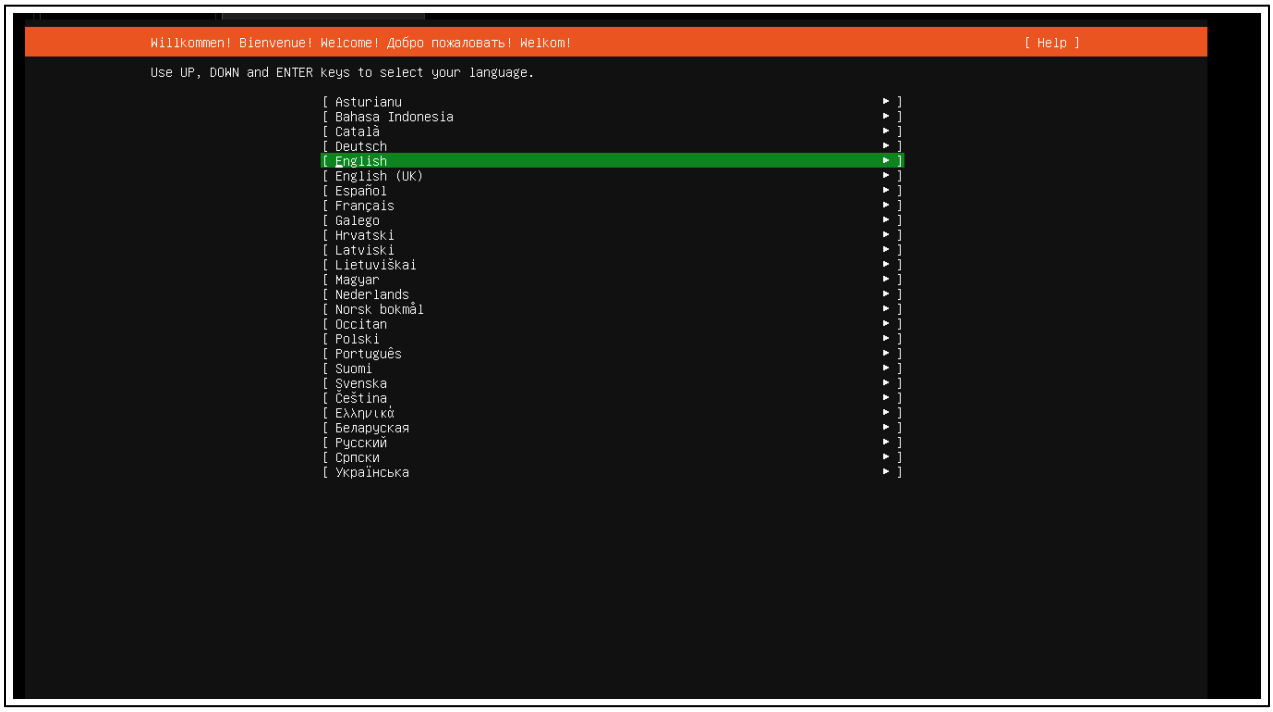


Figure 8.1: Ubuntu Server installer language menu with English selected.

Task 9: Keyboard Configuration

Explanation: The installer needs to know the keyboard layout so that characters typed during and after installation are correct. I set both the Layout and the Variant to English (US), which matches my physical keyboard.

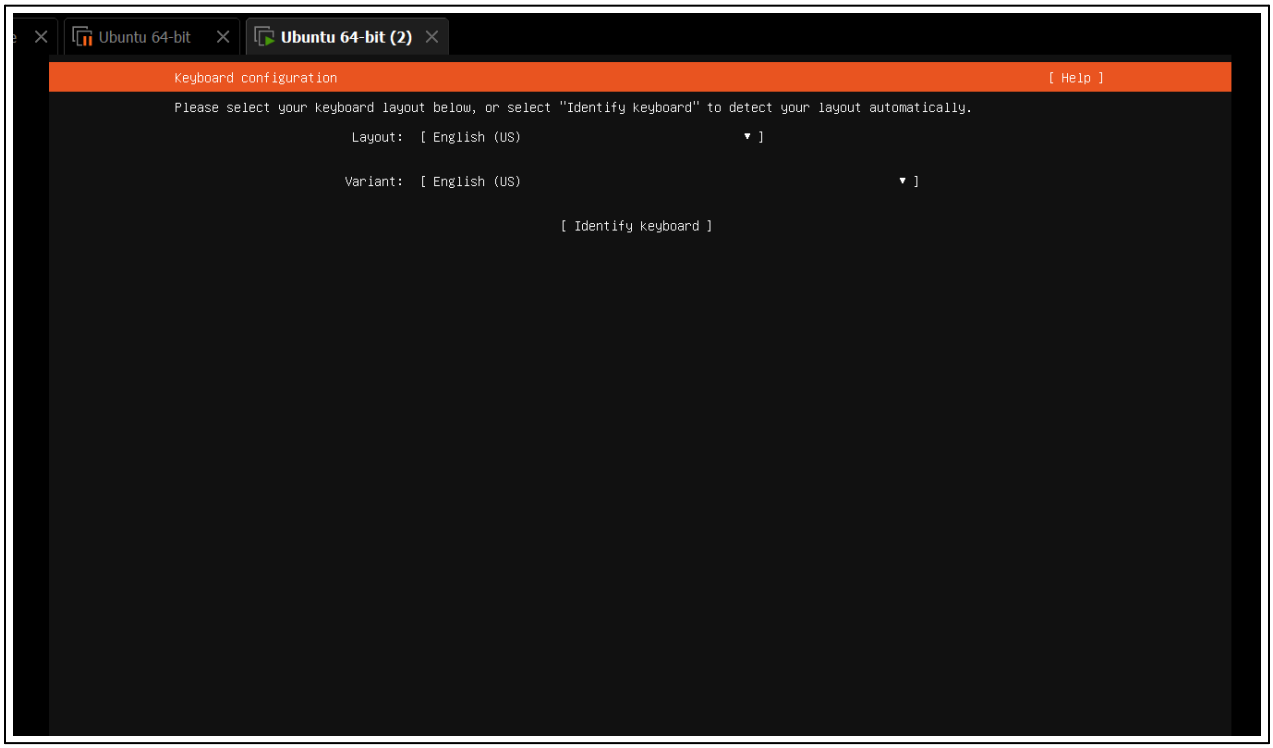


Figure 9.1: Keyboard configuration screen with layout and variant both set to English (US) (covers keyboard layout and variant evidence).

Task 10: Choose the Installation Type

Explanation: The installer offers a standard Ubuntu Server installation and a minimized one. I kept the default Ubuntu Server option, because it includes the packages normally needed to operate and administer a server. I did not enable third-party drivers.

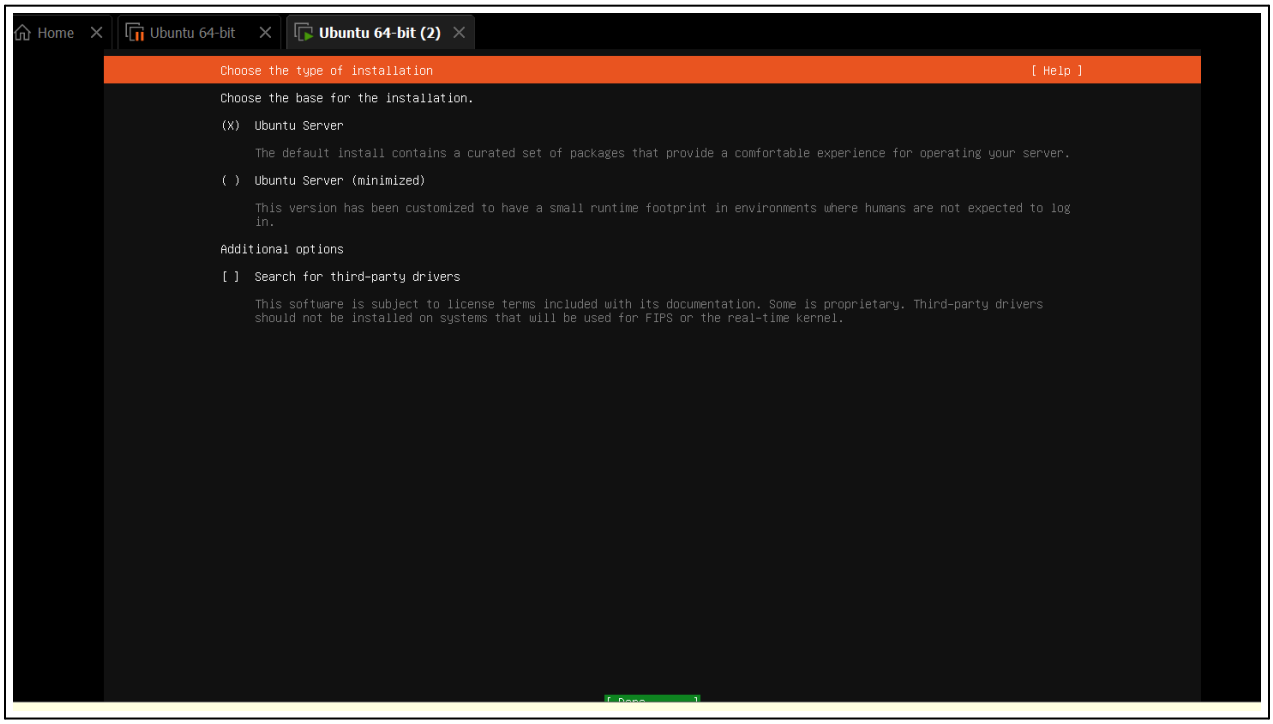


Figure 10.1: Installation type screen with the default Ubuntu Server option selected.

Task 11: Network Configuration

Explanation: A server needs at least one working network interface in order to receive updates and to accept remote SSH connections. The installer detected the interface ens33, which is the virtual Intel network adapter, and configured it automatically with DHCP, giving it an address in the 192.168.82.x subnet. I accepted this automatic configuration.

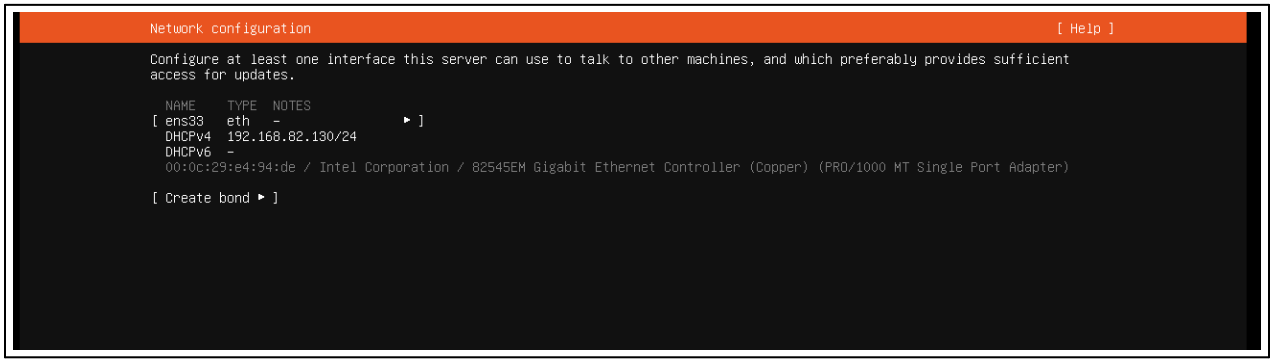


Figure 11.1: Network configuration showing interface ens33 with an IPv4 address assigned by DHCP in the 192.168.82.x subnet (covers installer network and network interface evidence).

Task 12: Proxy and Mirror Configuration

Explanation: A proxy is only needed when the network requires one to reach the internet, so I left the proxy address blank. The installer also asks for an Ubuntu archive mirror, which is the server that provides packages and updates. I kept the default mirror and the installer tested its location before continuing.



Figure 12.1: Proxy configuration screen with the proxy address left blank.

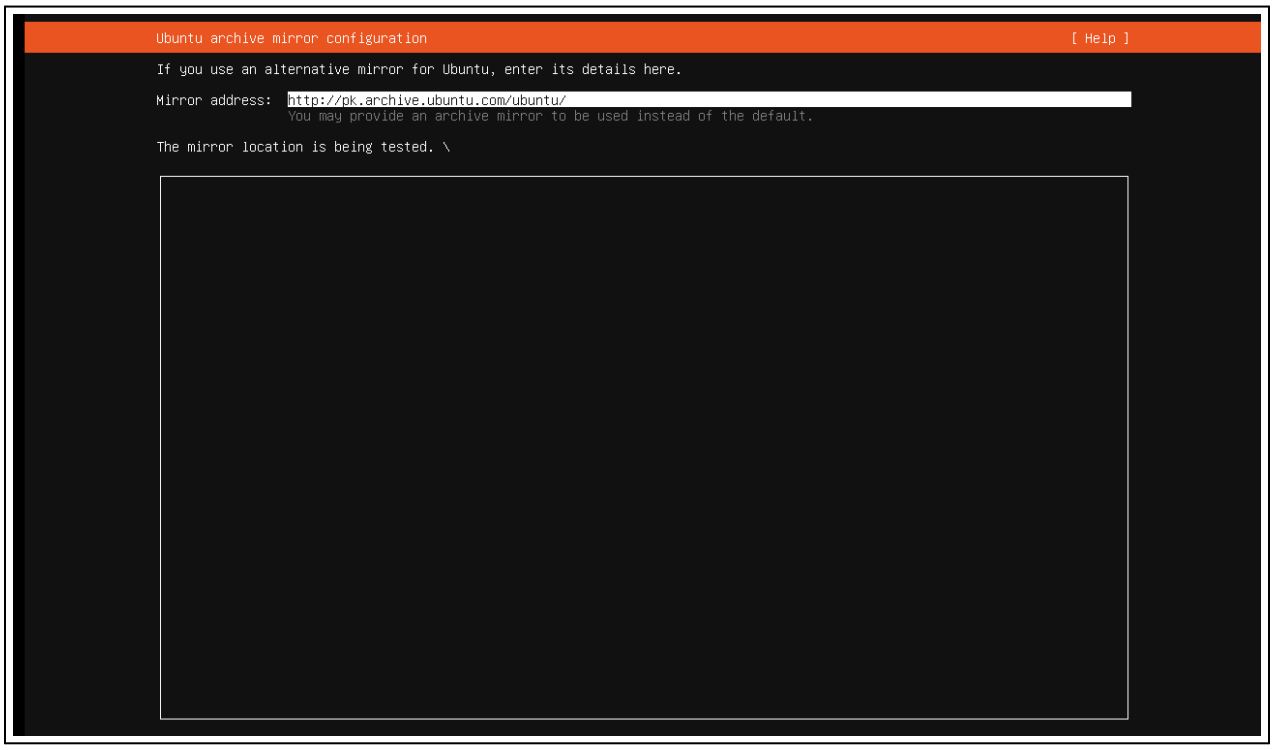


Figure 12.2: Archive mirror configuration screen showing the default mirror address, with the mirror location being tested.

Task 13: Guided Storage Configuration

Explanation: The installer needs to know how to partition the virtual disk. I chose the guided option Use an entire disk, selected the 20 GB disk /dev/sda, and enabled Set up this disk as an LVM group. LVM (Logical Volume Manager) makes it easier to resize volumes later. I did not encrypt the LVM group with LUKS.

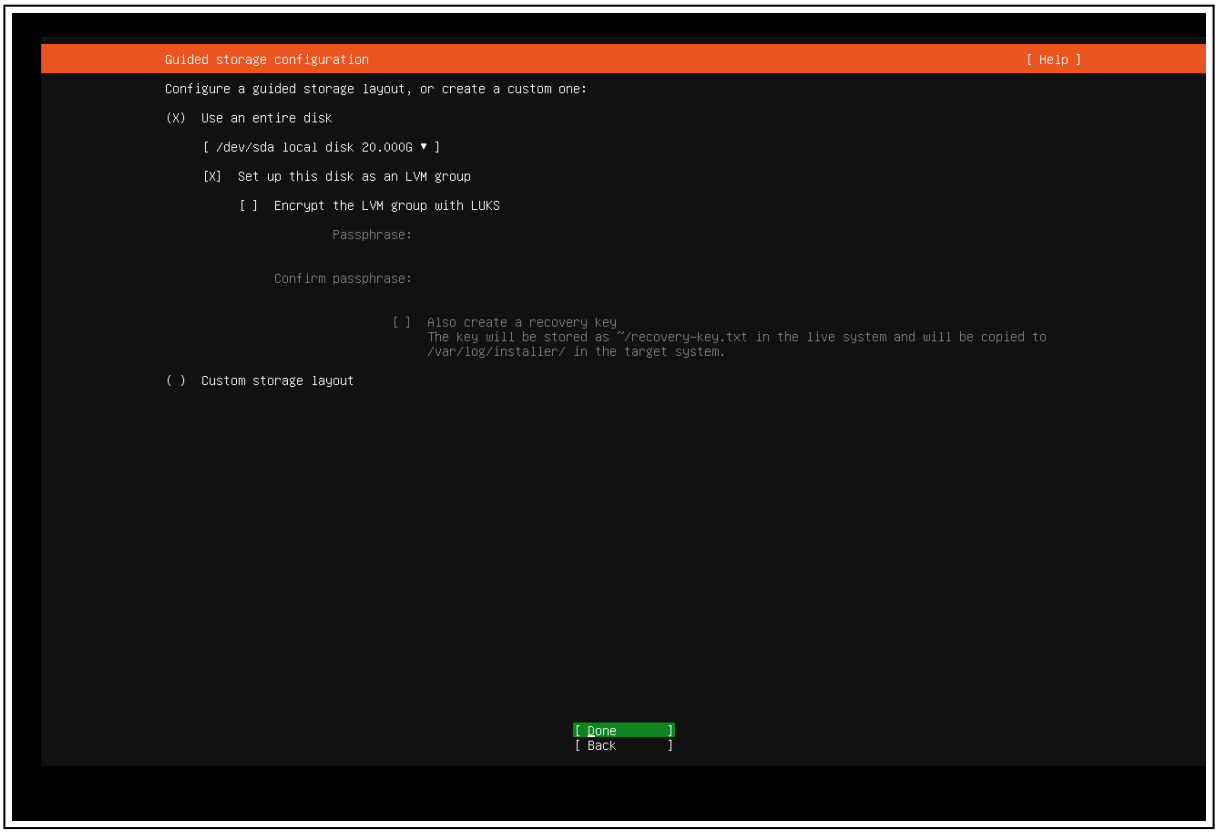


Figure 13.1: Guided storage configuration using an entire 20 GB disk (/dev/sda) as an LVM group, with LUKS encryption not enabled.

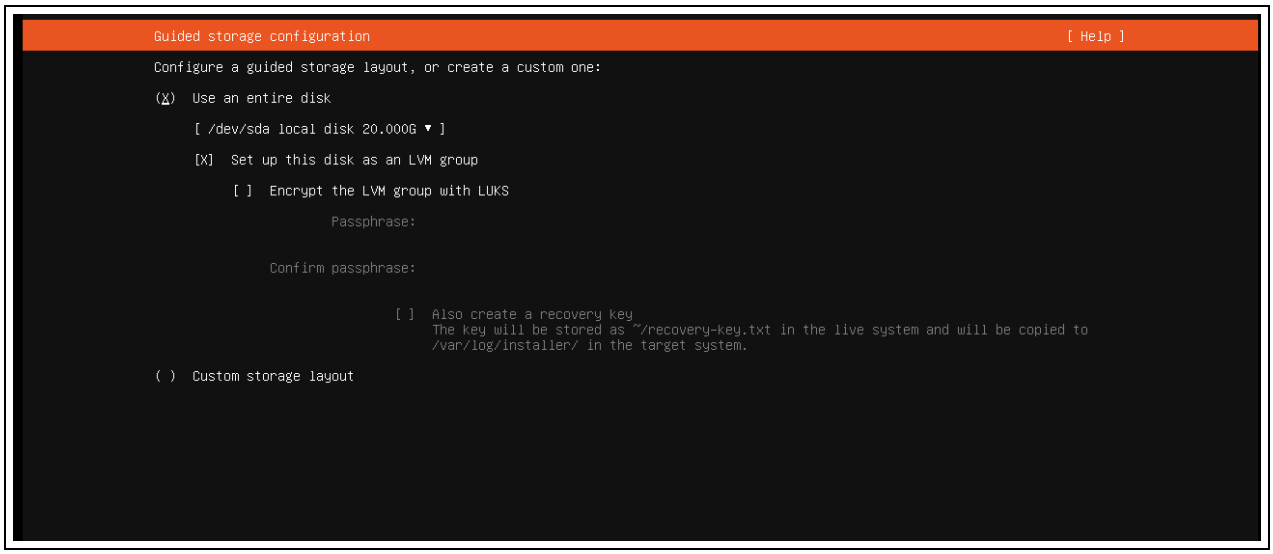


Figure 13.2: Guided storage configuration screen again showing the entire disk /dev/sda set up as an LVM group.

Task 14: Storage Summary and Confirmation

Explanation: Before anything is written to disk, the installer presents the layout it will create. The summary lists a 10 GB ext4 root volume (/) on LVM and a 1.771 GB ext4 /boot partition on /dev/sda, together with a small BIOS boot spacer. I reviewed it and chose Done. A final Confirm destructive action prompt warns that the selected disk will be formatted, and I confirmed with Continue because the virtual disk is new and empty.

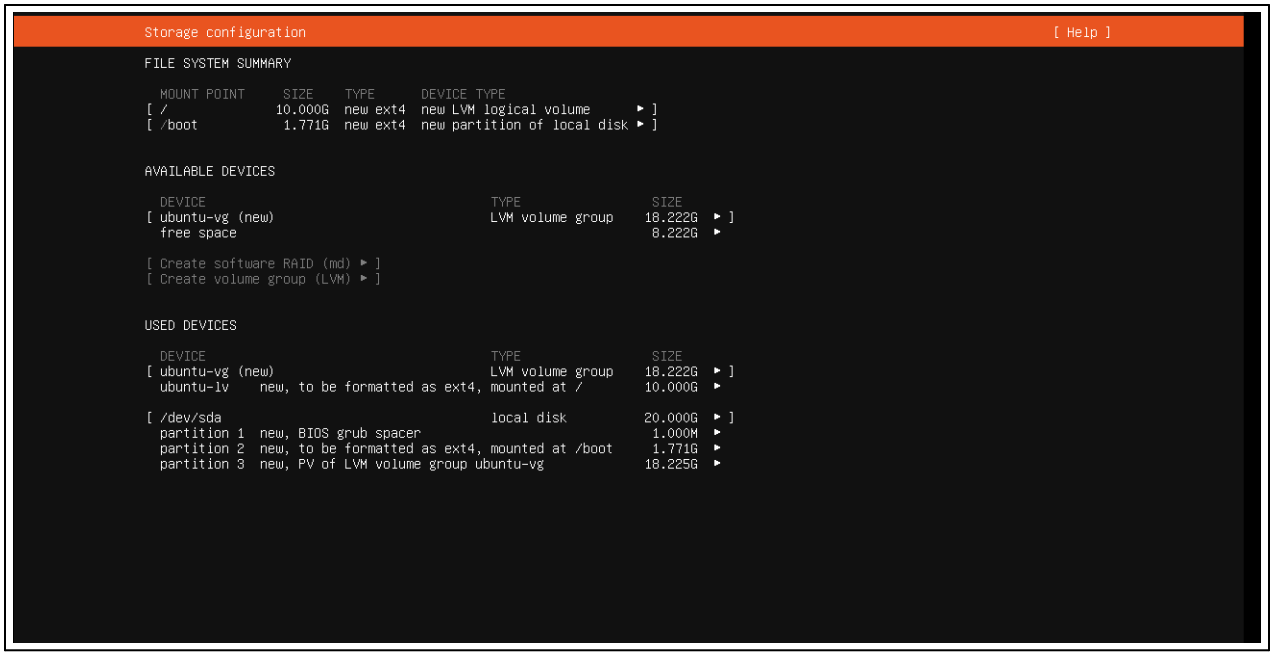


Figure 14.1: Storage summary showing / (10 GB ext4 on LVM) and /boot (1.771 GB ext4) on /dev/sda (covers storage configuration, partition layout, selected disk, and disk details evidence).

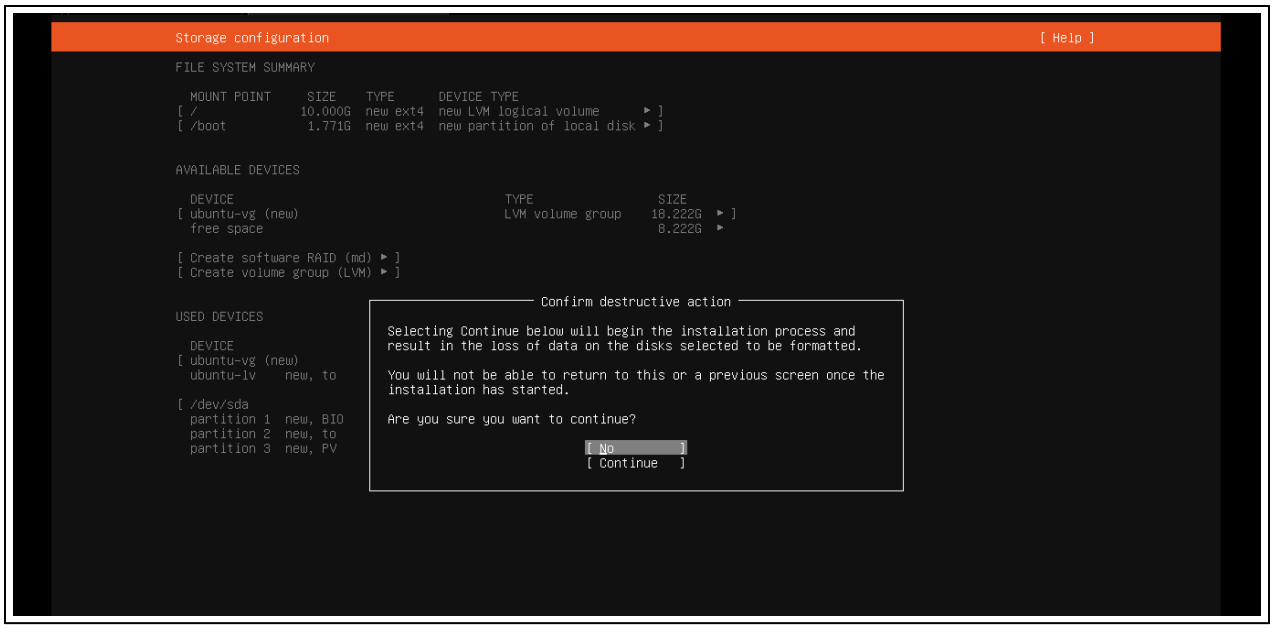


Figure 14.2: Confirm destructive action dialog before the installation starts (covers storage warning and storage confirmed evidence).

Task 15: Profile Setup: Name, Server Name and Username

Explanation: This screen creates the administrator account. I entered my full name Eman Fatima, set the server name to ubuntu, and picked the username eman-fatima-7125, which is my GitHub username written in lowercase because Ubuntu usernames must be lowercase. I also set a password, which the installer masks on screen. Using these values makes the terminal prompt show eman-fatima-7125@ubuntu.

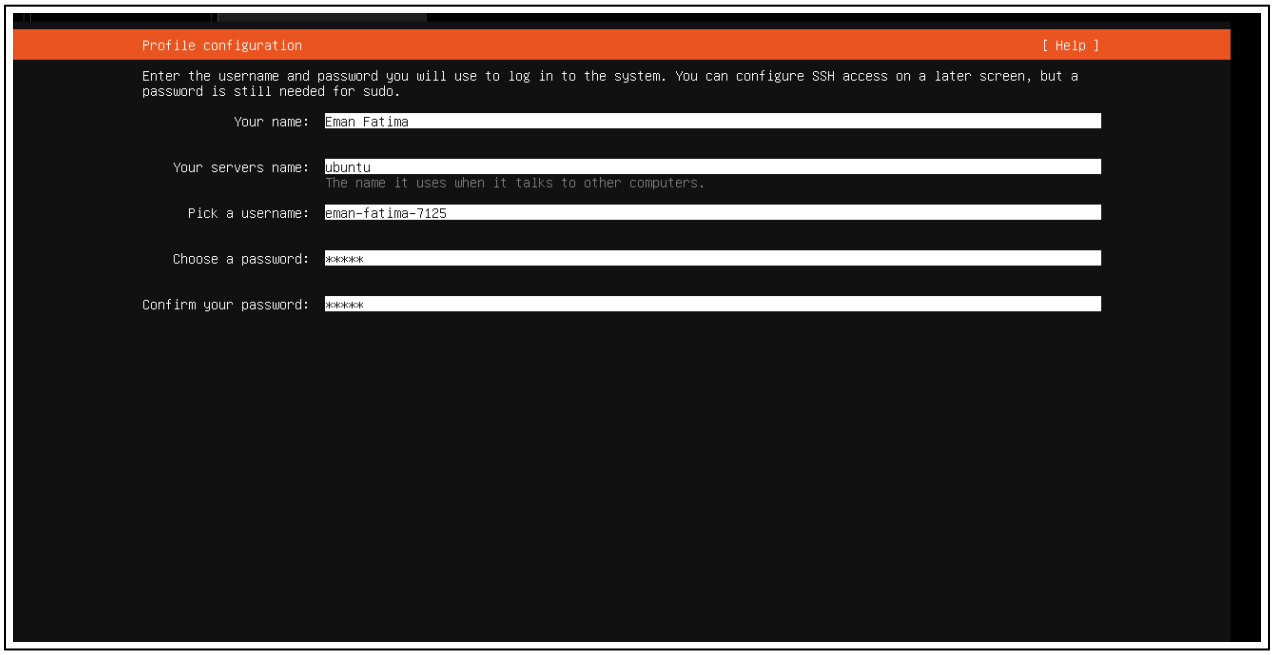


Figure 15.1: Profile configuration with name Eman Fatima, server name ubuntu, and username eman-fatima-7125; the password is masked (covers profile setup, actual name, server name, and username evidence).

Task 16: Ubuntu Pro, SSH, Snaps and Installation

Explanation: The remaining screens ask for optional services. I skipped Ubuntu Pro for now, since it is not needed for this lab. The SSH configuration screen is where the OpenSSH server is offered, which is required for the remote login in Task 17. The featured snaps screen lists optional server software. The installer then wrote the system to disk, and the log finished with an Installation complete message and a Reboot Now option.

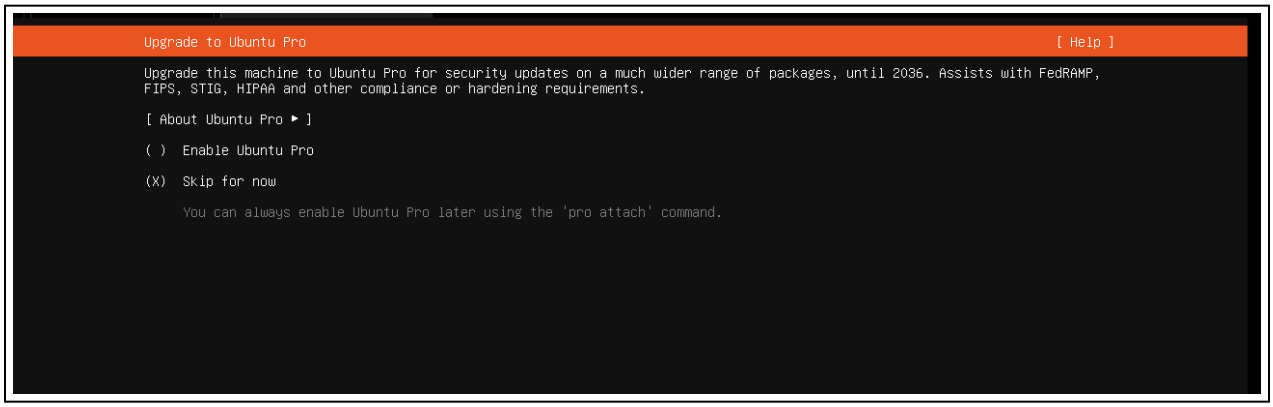


Figure 16.1: Upgrade to Ubuntu Pro screen with Skip for now selected.

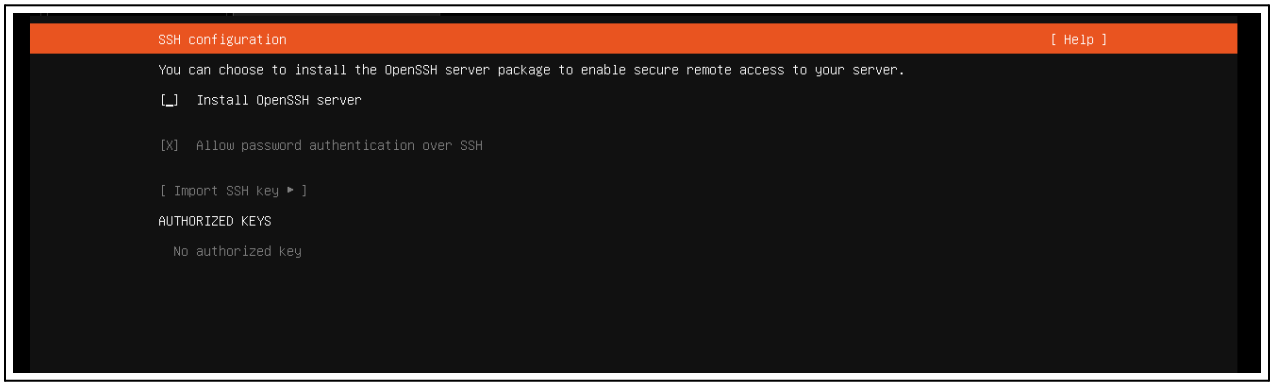


Figure 16.2: SSH configuration screen offering the OpenSSH server option used for remote access.



Figure 16.3: Featured server snaps screen shown during setup.

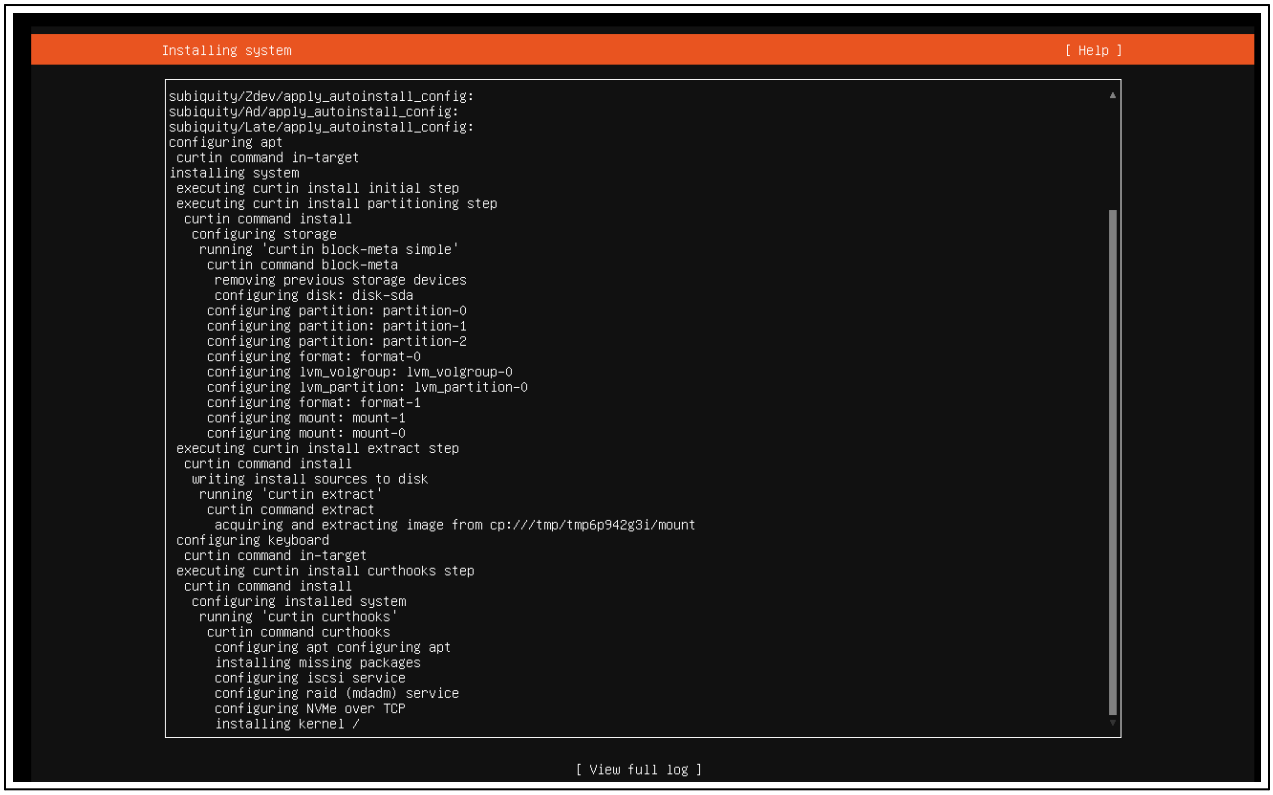


Figure 16.4: Installing system progress log (covers installation progress and installer boot evidence).

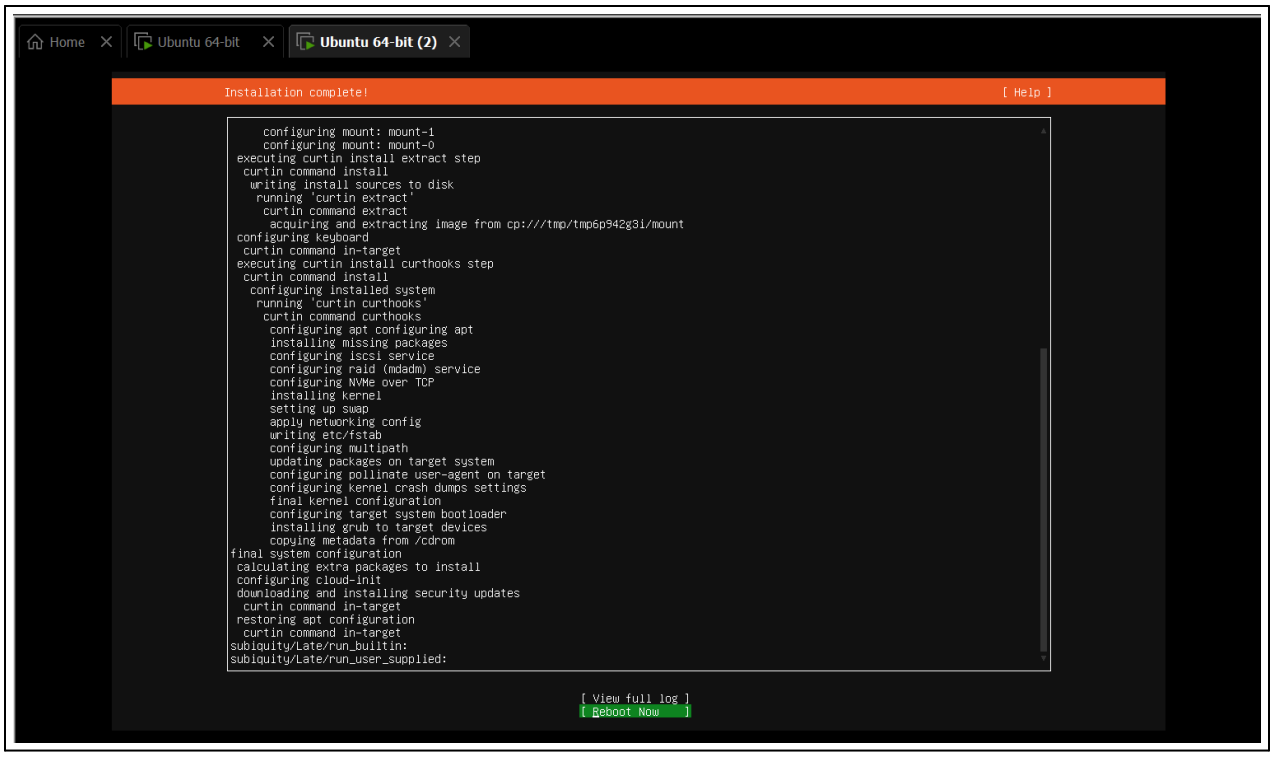


Figure 16.5: Installation complete screen with the Reboot Now option inside VMware Workstation.

Task 17: Find the IP Address and Connect over SSH

Explanation: After rebooting, I logged in at the VMware console and ran the ip addr command to find the server address, which is 192.168.82.131 on interface ens33. From Windows PowerShell I connected with ssh eman-fatima-7125@192.168.82.131. On the first connection SSH showed the host key fingerprint, and I accepted it by typing yes. After I entered the password, the Ubuntu 26.04.1 LTS welcome banner appeared and the prompt showed eman-fatima-7125@ubuntu, which confirms the remote session works.

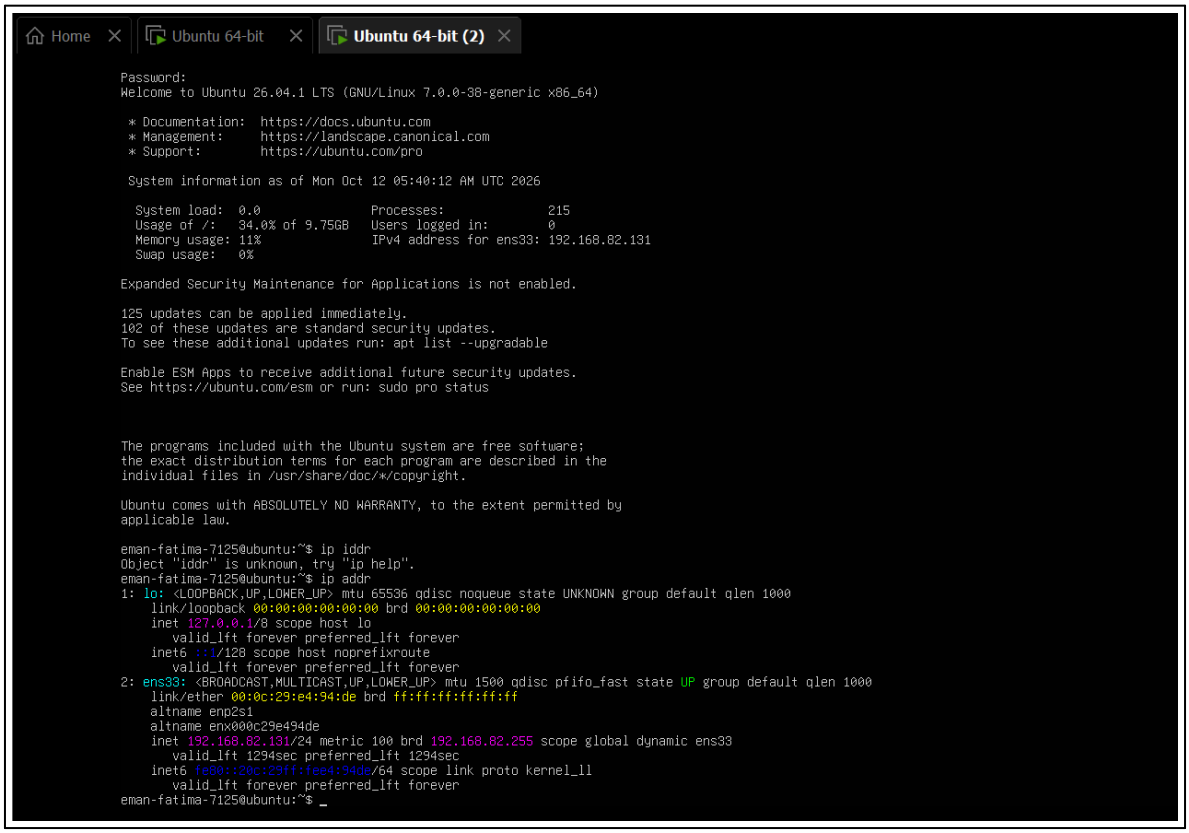


Figure 17.1: Local Ubuntu terminal with the prompt eman-fatima-7125@ubuntu and the ip addr output showing IPv4 address 192.168.82.131 on ens33.

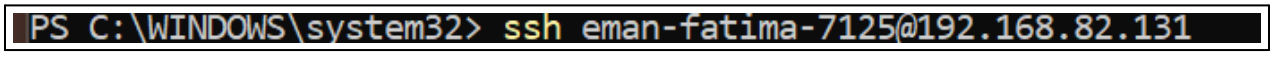


Figure 17.2: SSH command run in Windows PowerShell: ssh eman-fatima-7125@192.168.82.131.

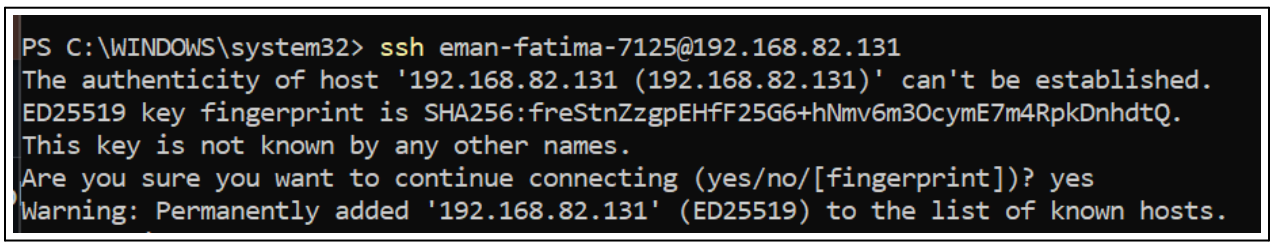


Figure 17.3: First-connection host key prompt, accepted by typing yes, which adds the server to known hosts.

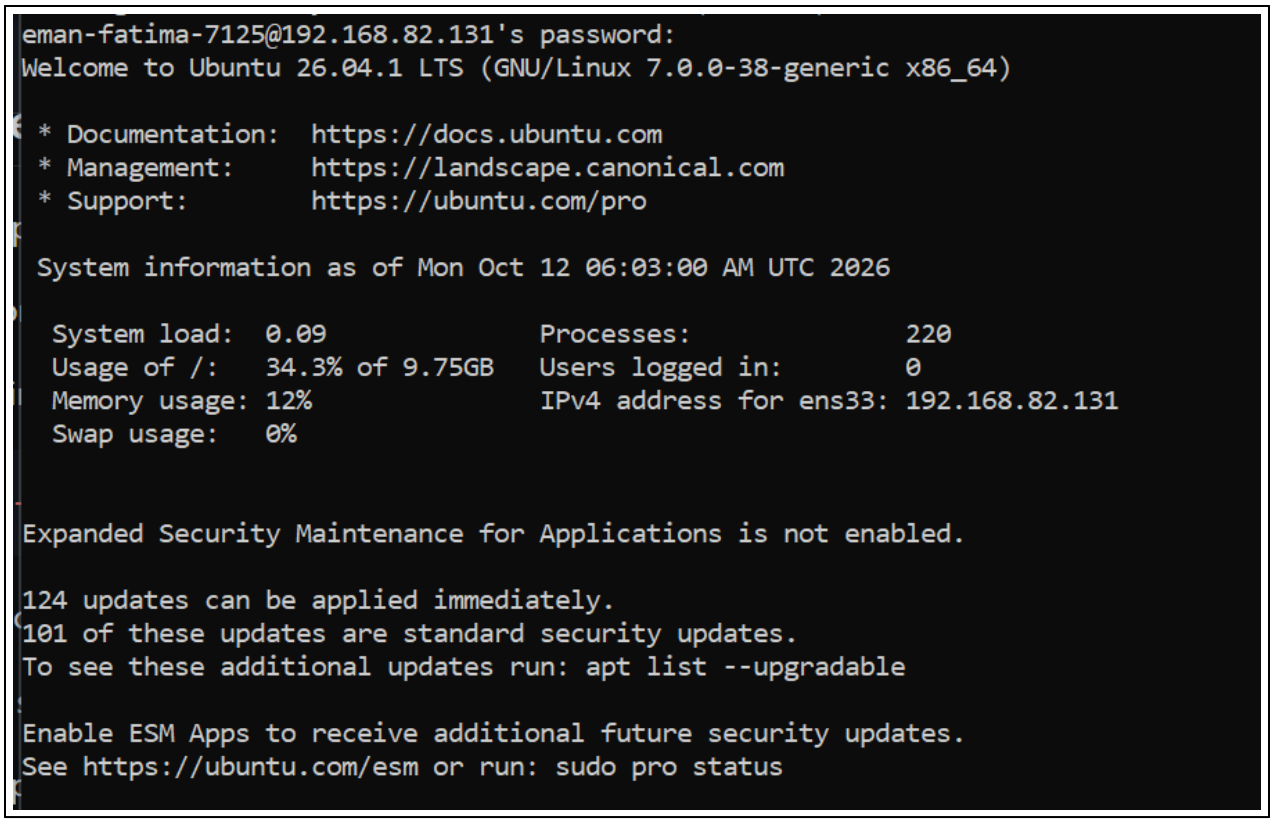


Figure 17.4: Successful SSH login with the Ubuntu 26.04.1 LTS welcome banner and IPv4 address 192.168.82.131.

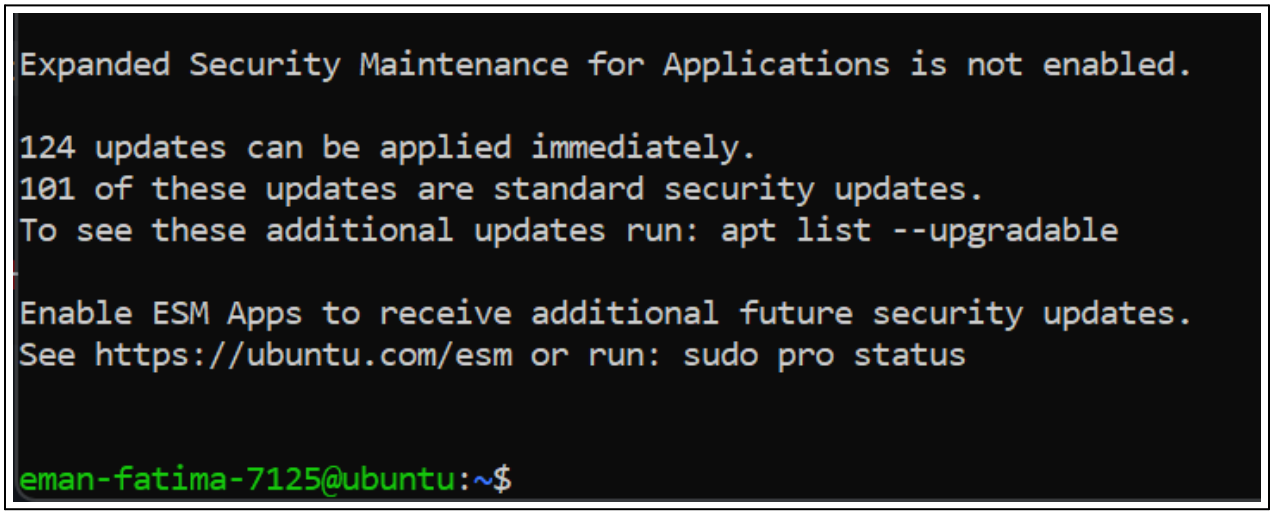


Figure 17.5: Remote SSH prompt showing eman-fatima-7125@ubuntu, confirming the connection to the Ubuntu server.